

7.3 How can materials and processes be used to make commercial products?

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| <p>a. Understand commercial production processes and machinery used to manufacture products to different scales of production, including:</p> <ul style="list-style-type: none">i. moulding methods, such as injection, rotational, compression, extrusion and blowii. thermoforming and vacuum formingiii. die casting and sand castingiv. sheet metal forming and stampingv. automated material handling systemsvi. robotic arms to stack, assemble, join and paint parts. |  |
| <p>b. Understand how the design of jigs, fixtures, presses, formers and moulds in commercial production are used to ensure consistent accuracy and quality, and different scales of production methods.</p> |  |
| <p>c. Understand the necessity for manufacturers to optimise the use of materials and production processes, such as:</p> <ul style="list-style-type: none">• economical lay plans and costing; ensuring cost effective production for viability• working to a budget through efficient manufacture and making the best use of labour and capital throughout the design and manufacturing process. |  |